THE UNIVERSITY OF MELBOURNE

Student Number

Semester 1 Assessment, 2020

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

This exam consists of 18 pages (including this page)

Authorised materials: printed one-sided copy of the Exam or the Masked Exam made available earlier (or an offline electronic PDF reader), handwritten notes, and blank A4 paper

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Instructions to Students

- During exam writing time you may only interact with the device running the Zoom session with supervisor permission. The screen of any other device must be visible in Zoom from the start of the session.
- If you have a printer, print out the exam single-sided and hand write your solutions into the answer spaces.
- If you do not have a printer, or if your printer fails on the day of the exam,

  (a) download the exam paper to a second device (not running Zoom), disconnect it from the internet as soon as the paper is downloaded and read the paper on the second device;

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- If you are unable to answer the whole question in the answer space provided then you can append additional handwritten solutions to the end of your exam submission. If you do this you MUST make a note in the correct answer space or page for the question, warning the marker that you have appended additional remarks at the end.
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- Confirm with your Zoom supervisor that you have GradeScope confirmation of submission before leaving Zoom supervision.
- You should attempt all questions.
- There are 10 questions with marks as shown. The total number of marks available is 100.

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Question 1 (27 marks)

Short-answer questions: Full marks will be given for a correct answer placed in the box. Part marks will be given for incorrect answers if working is shown.

(a) Evaluate $\cos\left(-\frac{\pi}{3}\right) + \sin\left(\frac{7\pi}{2}\right)$

Work out each term separately

$$\cos\left(-\frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \quad \boxed{1}$$

$$\sin\left(\frac{7\pi}{2}\right) = \sin\left(4\pi - \frac{\pi}{2}\right) = \sin\left(-\frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2}\right) = -1 \quad \boxed{1}$$

Therefore $\cos\left(-\frac{\pi}{3}\right) + \sin\left(\frac{7\pi}{2}\right) = \frac{1}{2} - 1 = -\frac{1}{2}$ $\boxed{1\,\text{A}}$

Subtotal 3

(b) An Italian restaurant is offering a special degustation deal for large groups. They offer any combination of 5 different pizzas and 5 different pastas for \$150. Their menu has 15 different pizzas and 10 different pasta options to choose from. How many different degustation menus can the customers make up?

In this case, order doesn't matter. There are 15 pizzas to choose from, giving $\binom{15}{5} = \frac{15!}{5!10!} = \frac{15\cdot14\cdot13\cdot12\cdot11\cdot10!}{10!\cdot5\cdot4\cdot3\cdot2} = 3003$ ways to choose the 5 pizzas in the degustation. Similarly, there are 10 pastas to choose from, giving $\binom{10}{5} = \frac{10!}{5!5!} = \frac{10\cdot9\cdot8\cdot7\cdot6\cdot5!}{5!\cdot5\cdot4\cdot3\cdot2} = 252$ ways to choose the 5 pastas in the degustation. Since these options can be chosen independently, we have a total of:

$$\binom{15}{5} \times \binom{10}{5} = 3003 \times 252 = 756756 \quad \text{options} \quad \boxed{1\,\text{A}}$$

$\boxed{1\,\text{M}}$(combinations) $\boxed{1\,\text{M}}$ (multiplication principle)

Subtotal 3

(c) Let $f(x) = (x^2 + 4x + 2)\sin(x)$. Find $f'(x)$.

Use product rule $f = u \cdot v$:

$$u = x^2 + 4x + 2 \qquad u' = 2x + 4$$
$$v = \sin(x) \qquad v' = \cos(x)$$

So $f'(x) = (2x + 4)\sin(x) + (x^2 + 4x + 2)\cos(x)$ $\boxed{1\,\text{A}}$

$\boxed{1\,\text{M}}$(product rule) $\boxed{1}$ (correct u' and v')

Subtotal 3

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(d) Let $g(x) = \dfrac{e^x}{\sqrt{x}}$. Find $g'(x)$.

Use quotient rule $g = \frac{u}{v}$:

$$u = e^x \qquad u' = e^x$$

$$v = \sqrt{x} \qquad v' = \frac{1}{2\sqrt{x}}$$

So $g'(x) = \dfrac{e^x \cdot \sqrt{x} - e^x\left(\frac{1}{2\sqrt{x}}\right)}{(\sqrt{x})^2} = \dfrac{e^x}{x}\left(\sqrt{x} - \dfrac{1}{2\sqrt{x}}\right)$ $\boxed{1\,\text{A}}$

$\boxed{1\,\text{M}}$(quotient rule) $\boxed{1}$ (correct u' and v')

$\boxed{\text{Subtotal 3}}$

(e) Let $h(x) = (\log(6x))^3$. Find $h'(x)$.

We need two chain rules:

$$h(a) = a^3 \qquad h'(a) = 3a^2$$

$$a(b) = \log(b) \qquad a'(b) = \frac{1}{b}$$

$$b(x) = 6x \qquad b'(x) = 6$$

So $h'(x) = 3(\log(6x))^2 \cdot \dfrac{1}{6x} \cdot 6 = \dfrac{3(\log(6x))^2}{x}$ $\boxed{1\,\text{A}}$

$\boxed{1\,\text{M}}$(chain rule) $\boxed{1}$ (correct h', a' and b')

$\boxed{\text{Subtotal 3}}$

(f) Find the equation of the tangent line of $f(x) = (\cos(x) + 1)^2$ about $x = \frac{\pi}{2}$

Work out each component separately

$f(x) = (\cos(x) + 1)^2 \qquad f(\frac{\pi}{2}) = (\cos(\frac{\pi}{2}) + 1)^2 = 1$ $\boxed{1}$

$f'(x) = -2(\cos(x) + 1)\sin(x) \quad f'(\frac{\pi}{2}) = -2(\cos(\frac{\pi}{2}) + 1)\sin(\frac{\pi}{2}) = -2$ $\boxed{1}$

So the tangent line of $f(x)$ about $x = \frac{\pi}{2}$ is:

$$y - f(\frac{\pi}{2}) = f'(\frac{\pi}{2})(x - \frac{\pi}{2})$$

$$y = f(\frac{\pi}{2}) + f'(\frac{\pi}{2})(x - \frac{\pi}{2})$$

$$y = 1 - 2(x - \frac{\pi}{2}) = -2x + \pi + 1$$ $\boxed{1\,\text{A}}$

$\boxed{\text{Subtotal 3}}$

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(g) Evaluate the following:

$$\int \frac{3x^5 + 2x}{x^2}\, dx$$

$$\int \frac{3x^5 + 2x}{x^2}\, dx = \int 3x^3 + \frac{2}{x}\, dx \quad \boxed{1\,\mathsf{M}}$$

Therefore, $\int \frac{3x^5+2x}{x^2}\, dx = \frac{3}{4}x^4 + 2\log|x| + C$ $\boxed{2\,\mathsf{A}}$ (one for each correct term)

$\boxed{\text{Subtotal 3}}$

(h) Evaluate the following:

$$\int_0^8 \frac{1}{\sqrt[3]{x^2}} + \cos(\pi x)\, dx$$

$$\begin{aligned}
\int_0^8 \frac{1}{\sqrt[3]{x^2}} + \cos(\pi x)\, dx &= \int_0^8 x^{-\frac{2}{3}} + \cos(\pi x)\, dx \\
&= \left[\frac{x^{\frac{1}{3}}}{\frac{1}{3}} + \frac{1}{\pi}\sin(\pi x)\right]_0^8 \quad \boxed{1} \\
&= \left[3(8)^{\frac{1}{3}} + \frac{1}{\pi}\sin(8\pi)\right] - \left[3(0)^{\frac{1}{3}} + \frac{1}{\pi}\sin(0)\right] \quad \boxed{1} \\
&= \left[3\sqrt[3]{8} + \frac{1}{\pi}\cdot 0\right] - \left[0 + \frac{1}{\pi}\cdot 0\right] \\
&= 3\dot{2} = 6 \quad \boxed{1\,\mathsf{A}}
\end{aligned}$$

(i) Solve the following initial value problem

$\boxed{\text{Subtotal 3}}$

$$F(x) = \int e^{-5x}\, dx \quad \text{if } F(0) = 0$$

$$\begin{aligned}
F(x) &= \int e^{-5x}\, dx = -\frac{1}{5}e^{-5x} + C \quad \boxed{1} \\
F(0) &= -\frac{1}{5}e^{-5\cdot 0} + C \quad \boxed{1\,\mathsf{M}} \\
0 &= -\frac{1}{5} + C \\
C &= \frac{1}{5}
\end{aligned}$$

Hence, $F(x) = -\frac{1}{5}e^{-5x} + \frac{1}{5}$. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 3}}$

Full-solution problems: Complete solutions are required for the remainder of this examination. Be sure to justify your answers and to show all your work. In addition to the accuracy of your answer and methods, you will be assessed on the justification of your solutions.

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Question 2 (7 marks)

Find all solutions of the trigonometric equation $2\cos(6x) + \sqrt{3} = 0$.

$$\begin{aligned} 2\cos(6x) + \sqrt{3} &= 0 \\ 2\cos(6x) &= -\sqrt{3} \\ \cos(6x) &= \frac{-\sqrt{3}}{2} \quad \boxed{1} \end{aligned}$$

This implies that the angle in the range $[0, 2\pi)$ is $\frac{5\pi}{6}$ or $\frac{7\pi}{6}$. $\boxed{2}$
Therefore, the general solution is:

$$\begin{aligned} 6x &= \frac{5\pi}{6} + 2k\pi, \frac{7\pi}{6} + 2k\pi, \qquad k \in \mathbb{Z} \quad \boxed{2\,\mathsf{M}} \\ x &= \frac{5\pi}{36} + \frac{k\pi}{3}, \frac{7\pi}{36} + \frac{k\pi}{3}, \qquad k \in \mathbb{Z} \end{aligned}$$

So

$$x \in \left\{ \frac{5\pi}{36} + \frac{k\pi}{3} \middle| k \in \mathbb{Z} \right\} \cup \left\{ \frac{7\pi}{36} + \frac{k\pi}{3} \middle| k \in \mathbb{Z} \right\} \quad \boxed{2\,\mathsf{A}}$$

or

$$x = \ldots, -\frac{7\pi}{36}, \frac{5\pi}{36}, \frac{17\pi}{36}, \ldots \quad \text{or} \quad x = \ldots, -\frac{5\pi}{36}, \frac{7\pi}{36}, \frac{19\pi}{36}, \ldots$$

$\boxed{\text{Subtotal } 7}$

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Question 3 (7 marks)

(a) Evaluate the following limit if it exists. Otherwise, write DNE and justify why the limit does not exist.

$$\lim_{x\to 2} \frac{x^3 - 9x^2 + 26x - 24}{x - 2}$$

At $x = 2$, numerator $= (2)^3 - 9(2)^2 + 26(2) - 24 = 0$ and denominator $= 2 - 2 = 0$. $\boxed{1}$

So we have to simplify

$$\begin{array}{r|l}
 & x^2 \quad - 7x + 12 \quad \boxed{1\,\mathsf{M}} \\
\hline
x - 2 & x^3 - 9x^2 + 26x - 24 \\
 & -x^3 + 2x^2 \\
\hline
 & -7x^2 + 26x \\
 & 7x^2 - 14x \\
\hline
 & 12x - 24 \\
 & -12x + 24 \\
\hline
 & 0
\end{array}$$

$$\begin{aligned}
\lim_{x\to 2} \frac{x^3 - 9x^2 + 26x - 24}{x - 2} &= \lim_{x\to 2} \frac{(x-2)(x^2 - 7x + 12)}{x - 2} \\
&= \lim_{x\to 2} (x^2 - 7x + 12) \ \boxed{1} \\
&= 2^2 - 7 \cdot 2 + 12 = 2. \ \boxed{1\,\mathsf{A}}
\end{aligned}$$

$\boxed{\text{Subtotal 4}}$

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(b) Use the limit definition of the derivative to find $f'(x)$ given $f(x) = \frac{1}{2x}$ for $x \neq 0$. No marks will be awarded for using any other method.

Use the limit definition:

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{\frac{1}{2(x+h)} - \frac{1}{2x}}{h} \quad \boxed{1\,\mathsf{M}} \\
&= \lim_{h\to 0} \frac{\frac{x-(x+h)}{2x(x+h)}}{h} \\
&= \lim_{h\to 0} \frac{\frac{-h}{2x(x+h)}}{h} \\
&= \lim_{h\to 0} \frac{-h}{2x(x+h)h} \\
&= \lim_{h\to 0} \frac{-1}{2x(x+h)} \\
&= \lim_{h\to 0} \frac{-1}{2x^2+2xh} \quad \boxed{1} \\
&= -\frac{1}{2x^2} \quad \boxed{1\,\mathsf{A}}
\end{aligned}$$

Subtotal 3

Question 4 (10 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = -x^2 + 8x - 15$.
Hint: You may use, without proof, that the function can also be written as

$$f(x) = -x^2 + 8x - 15 = (3-x)(x-5) = -(x-4)^2 + 1$$

(a) What is the domain, codomain, and range of f?

Domain $= \mathbb{R}$, Codomain$= \mathbb{R}$ $\boxed{1\,\mathsf{A}}$
Range $= (-\infty, 1]$ $\boxed{1\,\mathsf{A}}$

Subtotal 2

(b) Find the largest value q such that the function $f : (-\infty, q] \to \mathbb{R}$ given by

$$f(x) = -x^2 + 8x - 15$$

is injective.

We can use the turning point formula to find $q = 4$ $\boxed{1\,\mathsf{A}}$

Subtotal 1

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(c) Find the interval C such that $f : (-\infty, q] \to C$ given by

$$f(x) = -x^2 + 8x - 15$$

is surjective.

Pick C to be the range of $f(x)$. $C = (-\infty, 1]$ $\boxed{1\,\text{A}}$

$\boxed{\text{Subtotal 1}}$

(d) Based on your choice of q and C, explain why the function $f : (-\infty, q] \to C$ given by $f(x) = -x^2 + 8x - 15$ is invertible.

Based on our choice of q and C,

- f is injective from (b)
- f is surjective from (c) $\boxed{1}$

So f is bijective and hence invertible. $\boxed{1}$

$\boxed{\text{Subtotal 2}}$

(e) Define g to be the inverse of f. State the domain and codomain of g as well as provide the algebraic equation for $g(x)$.

domain of g: $(-\infty, 1]$ and codomain of g is $(-\infty, 4]$. $\boxed{1}$
The inverse is:

$$\begin{aligned} y &= -(x-4)^2 + 1 \\ y - 1 &= -(x-4)^2 \\ 1 - y &= (x-4)^2 \\ -\sqrt{1-y} &= x - 4 && \text{Negative root } \boxed{1} \\ 4 - \sqrt{1-y} &= x \ \boxed{1\,\text{M}} \end{aligned}$$

So the inverse is given by

$$g(x) = 4 - \sqrt{1-x} \ \boxed{1\,\text{A}}$$

$\boxed{\text{Subtotal 4}}$

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Question 5 (11 marks)

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = e^x(x^2 - 1)$.
Hint: It may be useful to know that $\sqrt{2} \approx 1.41$

(a) Find all x and y intercepts of the function.

- y-intercept at $f(0) = e^0(0^2 - 1) = -1$. So y-intercept is at $(0, -1)$ $\boxed{1\,\mathsf{A}}$
- x-intercept at $f(x) = 0$, so $0 = e^x(x^2 - 1) = e^x(x - 1)(x + 1)$. Solving for x gives $x = -1, 1$, since $e^x > 0$. So x-intercepts are at $(-1, 0)$ and $(1, 0)$. $\boxed{1\,\mathsf{A}}$

(b) Find the intervals where $f(x)$ is positive and intervals where $f(x)$ is negative

Subtotal 2

Partition domain by roots:

- $f(-2) > 0$ so $f(x)$ positive on $(-\infty, -1)$
- $f(0) < 0$ so $f(x)$ negative on $(-1, 1)$
- $f(2) > 0$ so $f(x)$ positive on $(1, \infty)$ $\boxed{1\,\mathsf{M}}$ $\boxed{1\,\mathsf{A}}$

(c) Evaluate and simplify $f'(x)$

Subtotal 2

Since $f(x) = e^x(x^2 - 1)$ Use product rule $f = u \cdot v$: $\boxed{1\,\mathsf{M}}$

$$u = e^x \qquad u' = e^x$$
$$v = x^2 - 1 \qquad v' = 2x$$

So we have:

$$\begin{aligned} f'(x) &= u'v + uv' = e^x(x^2 - 1) + e^x(2x) \\ &= e^x(x^2 + 2x - 1) \ \boxed{1\,\mathsf{A}} \\ &(= e^x(x - (-1 + \sqrt{2}))(x - (-1 - \sqrt{2}))) \end{aligned}$$

(d) Find the points in the domain where $f'(x) = 0$.

Subtotal 2

Find the roots of $x^2 + 2x - 1$ using the quadratic formula to obtain $f'(x) = e^x(x^2 + 2x - 1) = e^x(x - (-1 + \sqrt{2}))(x - (-1 - \sqrt{2}))$. $\boxed{1}$

Therefore $f'(x) = 0$ when $x = -1 + \sqrt{2}, -1 - \sqrt{2})$ since $e^x > 0$ for all x. $\boxed{1\,\mathsf{A}}$

Subtotal 2

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(e) Find the intervals where $f(x)$ is increasing and intervals where $f(x)$ is decreasing

Check the sign of the derivative on the different intervals:

- $f'(-3) = e^{-3}(-3+1-\sqrt{2}))(-3+1+\sqrt{2})) > 0$ so $f(x)$ is increasing on $(-\infty, -1-\sqrt{2})$
- $f'(0) = e^{0}(0+1-\sqrt{2}))(0+1+\sqrt{2})) < 0$ so $f(x)$ is decreasing on $(-1-\sqrt{2}, -1+\sqrt{2})$
- $f'(1) = e^{1}(1+1-\sqrt{2}))(1+1+\sqrt{2})) > 0$ so $f(x)$ is increasing on $(-1+\sqrt{2}, \infty)$ 1 M 1 A

Subtotal 2

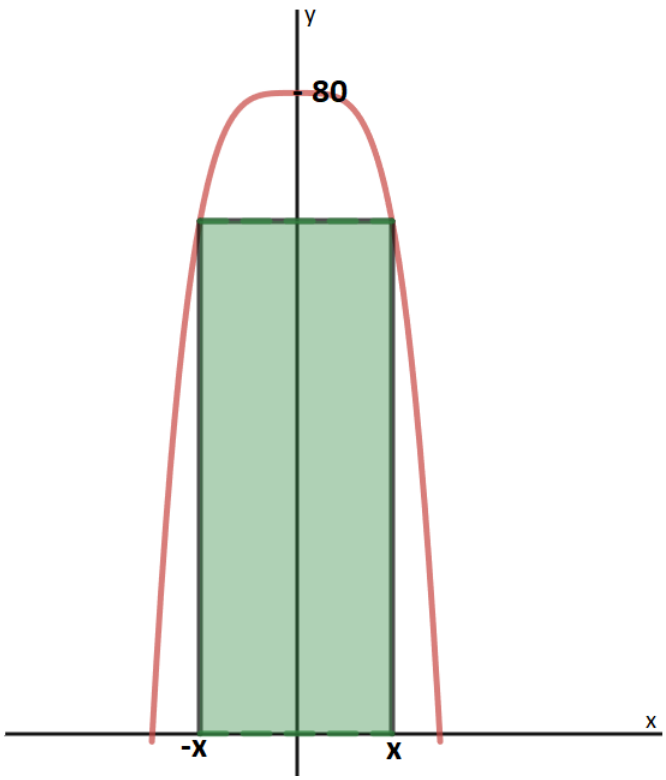
(f) Find and classify the location of any local extrema of the function.

There is a local maximum at $x = -1-\sqrt{2}$ and a local minimum at $x = -1+\sqrt{2}$ by the first derivative test. 1 A

Subtotal 1

Question 6 (11 marks)

The following graph shows the function $y = -x^4 + 80$ with a general rectangle drawn with its base along the x axis



Hint: You might find useful to know that $\sqrt[4]{80} \approx 2.99$ and $\sqrt[4]{16} = 2$.

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(a) What is the length of the base of the rectangle in terms of x?

The base is from $-x$ to x so its length is $2x$ $\boxed{1\,\text{A}}$

$\boxed{\text{Subtotal 1}}$

(b) Given that the upper vertices of the rectangle touch the curve, what is the height of the rectangle in terms of x?

The y value for x (or $-x$) is $-x^4 + 80$ so this is the height of the rectangle. $\boxed{1\,\text{A}}$

$\boxed{\text{Subtotal 1}}$

(c) Write down an expression for the Area of the rectangle in terms of x. What is the domain of $A(x)$.

Area = length $\times$ height $\boxed{1}$

So $\quad A = 2x(-x^4 + 80) = -2x^5 + 160x$. $\boxed{1\,\text{A}}$

The $x-$intercepts for $y = -x^4 + 80$ are $\pm\sqrt[4]{80} \approx \pm 2.99$. $\boxed{1}$

Therefore, the domain for x is $x \in [0, \sqrt[4]{80}]$. $\boxed{1\,\text{A}}$

$\boxed{\text{Subtotal 4}}$

(d) Determine the dimensions (length, height and area) of the rectangle with the maximum area that will fit under this curve.

For the maximum area we want to find the critical points.

Since A is a polynomial, we only need to find the points for which $A' = 0$. $\boxed{1\,\text{M}}$ That is, $-10x^4 + 160 = 0$. $\boxed{1}$(correct derivative)

Hence we want $x^4 = 16$ or $x = \pm 2$. Given the domain, we only take $x = 2$ $\boxed{1}$.

To show this is the max, we use the closed interval method. $\boxed{1\,\text{M}}$

$$\begin{aligned} A(0) &= 0, \\ A(2) &= -2(2)^5 + 160(2) = -64 + 320 = 256, \\ A(\sqrt[4]{80}) &= 0 \end{aligned}$$

Therefore, we have a maximum at $x = 2$. The length is 4 units, the height is 64 units and the Area is 256 square units. $\boxed{1\,\text{A}}$

$\boxed{\text{Subtotal 5}}$

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Question 7 (10 marks)

Consider two functions given by

$$y = x^2 + 1 \qquad \text{and} \qquad y = x + 3$$

(a) Find the coordinates where the two functions intersect.

The functions intersect when

$$x^2 + 1 = x + 3 \ \boxed{1\,\mathsf{M}}$$
$$x^2 - x - 2 = 0$$
$$(x-2)(x+1) = 0 \ \boxed{1}$$

So the intersections are at $x = 2, -1$. The points are $(2, 5)$ and $(-1, 2)$. $\boxed{2\,\mathsf{A}}$

$\boxed{\text{Subtotal 4}}$

(b) Find the **physical** area between the functions f and g.

Based on previous part, we only have to worry about the domain $[-1, 2]$.

$$A = \int_{-1}^{2} (x+3) - (x^2+1)dx \ \boxed{1\,\mathsf{M}} \ \boxed{1}\text{(right choice upper/lower function)}$$
$$= \int_{-1}^{2} -x^2 + x + 2dx$$
$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x\right]_{-1}^{2} \ \boxed{2}$$
$$= \left[-\frac{(2)^3}{3} + \frac{(2)^2}{2} + 2(2)\right] - \left[-\frac{(-1)^3}{3} + \frac{(-1)^2}{2} + 2(-1)\right] \ \boxed{1\,\mathsf{M}}$$
$$= \left[-\frac{8}{3} + 2 + 4\right] - \left[\frac{(1}{3} + \frac{1}{2} - 2\right]$$
$$= 5 - \frac{1}{2}$$
$$= \frac{9}{2}$$

The area is $\frac{9}{2}$. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 6}}$

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Question 8 (5 marks)

Consider the discrete probability distribution for the random variable X shown in the table below:

x	-1	0	1	2
$P(X=x)$	0.1	a	0.4	0.3

(a) Find the value of a

We know that $\sum_x P(X=x) = 1$. Therefore, we must have

$0.1+a+0.4+0.3 = 1$, which gives $a = 1-0.3-0.4-0.1 = 0.2$ 1 A

Subtotal 1

(b) Find $\mathbb{E}(X)$

To compute the expected value:

$$\begin{aligned}\mathbb{E}(X) &= (-1)P(X=-1) + (0)P(X=0) + (1)P(X=1) + (2)P(X=2)\\ &= (-1)(0.1) + (0)(0.2) + (1)(0.4) + (2)(0.3)\\ &= -0.1 + 0.4 + 0.6 = 0.9 \quad \text{1 A} \quad \text{1 M}\end{aligned}$$

Subtotal 2

(c) Find the probability that $X = 2$, given that $X \geq 1$

This is a conditional probability question

$$\begin{aligned}P(X=2|X\geq 1) &= \frac{P(X=2 \text{ and } X\geq 1)}{P(X\geq 1)}\\ &= \frac{P(X=2)}{P(X\geq 1)} \qquad \text{Since } X=2 \text{ implies } X\geq 1\\ &= \frac{P(X=2)}{P(X=1)+P(X=2)}\\ &= \frac{0.3}{0.4+0.3}\\ &= \frac{0.3}{0.7} = \frac{3}{7} \quad \text{1 A} \quad \text{1 M}\end{aligned}$$

Subtotal 2

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Question 9 (7 marks)

The weight of Australian avocados is normally distributed with mean 200g and standard deviation 10g.

(a) Suppose that we sell Australian avocados at three different prices, depending on the size (small, medium or large). If the bottom 10% of avocados by weight are sold at the price of "small" avocados. What is the maximum weight of a "small" avocado? Your answer should be accurate to 1 decimal place.

Let X be the weight of an Australian avocado. We know that the mean weight is 200, with standard deviation 10, so $X \sim N(200, 10)$.

We want to find z such that $P(Z < z) = 0.1$. $\boxed{1}$ Using symmetry, we have $P(Z < z) = 1 - P(Z < -z)$. Hence we seek $P(Z < -z) = 0.9$. $\boxed{1}$ From the table, this gives $-z = 1.28$ and $z = -1.28$.

Converting to a weight gives:

$$\begin{aligned} -1.28 &= \frac{x - 200}{10} \quad \boxed{1\,\mathsf{M}} \\ -12.8 &= x - 200 \\ 200 - 12.8 &= x \end{aligned}$$

Hence the maximum weight of a "small" avocado is 187.2g. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 4}}$

(b) Avocados sold at the price of "large" avocados must weight at least 215g. What percentage of avocados can be sold at that price?

Covert to standard normal first:

We want $P(X \geq 215) = P(Z \geq \frac{215 - 200}{10}) = P(Z \geq 1.5)$ $\boxed{1\,\mathsf{M}}$.
Now

$$\begin{aligned} P(Z \geq 1.5) &= 1 - P(Z \leq 1.5) && \text{(probability of the complement)} \ \boxed{1\,\mathsf{M}} \\ &= 1 - 0.9332 && \text{(using the table for the standard normal)} \\ &= 0.0668 \end{aligned}$$

This means that 6.68% of the avocados can be sold at the price for "large" avocados. $\boxed{1\,\mathsf{A}}$

$\boxed{\text{Subtotal 3}}$

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Question 10 (5 marks)

Careful reasoning is required for this question. Even the right answer with insufficient justification will not score well in this question. When rolling two dice, find the probability of getting a greater number on the first die than the one on the second one, given that the sum should equal 8.

Let Ω be the sample space for rolling two dice.
For each die individually, there are 6 possible outcomes and, since the outcomes on each die are independent, we have that $|\Omega| = 6 \times 6 = 36$.

Let A be the event of getting a greater number on the first die than on the second, and let B be the event that the sum of the two dice is equal to 8.

This is a conditional probability problem. We are looking for

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \boxed{1\,\mathsf{M}}$$

We must now find the elements in $A \cap B$.
We first look at the elements in B. There are only 5 ways to get a sum of 8 when rolling two dice. We have $B = \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$. $\boxed{1}$

The only elements in B that are also in A (i.e. that also satisfy that the number on the first die is greater than the one on the second) are $(5,3)$ and $(6,2)$. Therefore, $A \cap B = \{(5,3),(6,2)\}$. $\boxed{1}$

This gives $|B| = 5$ and $|A \cap B| = 2$. Hence

$$P(A \cap B) = \frac{|A \cap B|}{|\Omega|} = \frac{2}{36}$$
$$P(B) = \frac{|B|}{|\Omega|} = \frac{5}{36} \quad \boxed{1}$$

So

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{36}}{\frac{5}{36}} = \frac{2}{5} \quad \boxed{1\,\mathsf{A}}$$

$\boxed{\text{Subtotal 5}}$

End of exam questions—Total Available Marks = 100

Turn the page for appended material

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Formula Sheet

- **Combinatorics:**

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$
$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

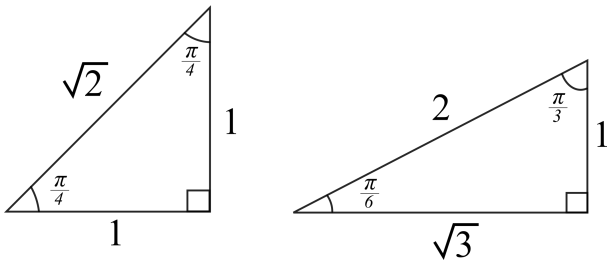
- **Functions:**
  - Lines

$$y = mx + c \qquad\qquad y - b = m(x-a)$$

  - Quadratics

$$y = ax^2 + bx + c = a(x-r)(x-s) = a(x-h)^2 + k$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Special Triangles:**



- **Trigonometric identities:**

$$\sin(x + 2\pi) = \sin(x) \qquad \cos(x + 2\pi) = \cos(x)$$
$$\sin(\pi - x) = \sin(x) \qquad \sin(-x) = -\sin(x)$$
$$\cos(\pi - x) = -\cos(x) \qquad \cos(-x) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$
$$\sin(2x) = 2\sin(x)\cos(x) \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$
$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \qquad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- **Exponential and logarithms:**

$$a^0 = 1 \qquad a^1 = a$$
$$a^{-1} = \frac{1}{a} \qquad a^{\frac{1}{2}} = \sqrt{a}$$
$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y}$$
$$(a^x)^y = a^{x \cdot y} \qquad \sqrt[y]{a^x} = a^{\frac{x}{y}}$$
$$a^x \cdot b^x = (ab)^x \qquad \frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

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$$\log_e(x) = \ln(x) \qquad \log_a(1) = 0$$
$$\log_a(a) = 1 \qquad \log_a(p^x) = x\log_a(p)$$
$$\log_a(pq) = \log_a(p) + \log_a(q) \qquad \log_a\left(\frac{p}{q}\right) = \log_a(p) - \log_a(q)$$
$$\log_a(p) = \frac{\log_c(p)}{\log_c(a)}$$

- **Limits:** If we have

$$\lim_{x\to a} f(x) = K \qquad \lim_{x\to a} g(x) = L$$

Then

$$\lim_{x\to a}[f(x)+g(x)] = K + L \qquad \lim_{x\to a}[f(x)-g(x)] = K - L$$
$$\lim_{x\to a}[f(x)\times g(x)] = K \times L \qquad \lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] = \frac{K}{L}$$
$$\lim_{x\to a}[f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \qquad \lim_{x\to a}[f(x)]^{\frac{1}{n}} = \left[\lim_{x\to a} f(x)\right]^{\frac{1}{n}}$$
$$\lim_{x\to a} cf(x) = cK$$
$$\lim_{x\to a} c = c \qquad \lim_{x\to a} x = a$$
$$\lim_{x\to a} x^n = a^n \qquad \lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}$$

- **Derivatives:**

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{a\to x}\frac{f(a)-f(x)}{a-x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(c\Big) = 0 \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(x^k\Big) = kx^{k-1}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\sqrt{x}\Big) = \frac{1}{2\sqrt{x}} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(e^x\Big) = e^x$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\log(x)\Big) = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\sin(x)\Big) = \cos(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\cos(x)\Big) = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\tan(x)\Big) = \frac{1}{\cos^2(x)}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(u(x)\cdot v(x)\Big) = \frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) + u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x} = u'(x)\cdot v(x) + u(x)\cdot v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u(x)}{v(x)}\right) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) - u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x}}{[v(x)^2]} = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)^2]}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(f(u(x))\Big) = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} = f'(u)\cdot u'(x)$$

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The standard normal distribution table for $P(X < x)$

x	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000